

## Additional Professional Training Datasets for Edge-AI PHM

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Here are additional, highly-regarded professional datasets that are recognized as global standards for benchmarking bearing and gearbox fault diagnosis. These complement the CWRU and NASA datasets and provide complex variables (variable speeds, noise, run-to-failure telemetry) crucial for training our Edge-AI model.

### 1. Paderborn University (PU) Bearing Dataset

**Dataset Access Link:** <https://mb.uni-paderborn.de/en/kat/main-research/datacenter/bearing-datacenter>

**Why this dataset (reason of importance and choosing):** The PU dataset is highly revered in the research community because it specifically focuses on variable operating conditions (varying speeds, load torques, and radial forces) and includes both artificially induced defects and natural run-to-failure faults. It is perfect for training our Edge AI to generalize and remain accurate even when the helicopter experiences sudden, dynamic aerodynamic loads (unlike strictly constant-speed datasets).

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### 2. IMS (Intelligent Maintenance Systems) / University of Cincinnati Bearing Dataset

**Dataset Access Link:** <https://ti.arc.nasa.gov/tech/dash/groups/pcoe/prognostic-data-repository/>

**Why this dataset (reason of importance and choosing):** Hosted alongside the NASA datasets, the IMS dataset provides high-frequency (20 kHz) continuous run-to-failure vibration data collected over several days until the bearings completely destroyed themselves (inner ring, outer ring, and roller element failures). This data is critical for prognostic modeling—allowing our TinyML model to not just detect a fault, but understand the sequential acoustic degradation leading up to a catastrophic failure.

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### 3. Southeast University (SEU) Planetary Gearbox Dataset

**Dataset Access Link:** <https://github.com/cathyshia/Mechanical-datasets> (Commonly accessed open-source repository)

**Why this dataset (reason of importance and choosing):** While CWRU and PU focus heavily on bearings, the SEU dataset specifically targets planetary gearboxes—the exact

mechanical component we are monitoring on the helicopter. It contains data for specific gear faults like root cracks, missing teeth, and surface wear under varying noisy conditions. Using this will allow our model to distinguish between a bearing hum and the specific high-frequency micro-harmonics of a fractured gear tooth.

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